Comprehensive Transcriptomic Analysis of Grapevine Red Blotch Virus (GRBV) Infection

A Multi-Dimensional RNA-Seq Study Across Infection Duration, Maturity Stages, Seasons, and Genomic Context

Data: 327 samples | 14,604 genes | 2 seasons | 4 maturity stages

25 figures | 6 analysis modules | 121 GWAS/QTL loci

Generated from the GRBV RNA-Seq Transcriptomics Analysis Pipeline

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1. Introduction

Grapevine Red Blotch Virus (GRBV) is a DNA virus (family Geminiviridae) that causes Red Blotch Disease (RBD) in grapevines, leading to reduced sugar accumulation, altered anthocyanin profiles, delayed ripening, and diminished fruit quality. Understanding the molecular mechanisms underlying GRBV infection is critical for developing management strategies, breeding resistant cultivars, and identifying diagnostic biomarkers.

This study presents a comprehensive RNA-Seq transcriptomic analysis of GRBV-infected (RB+) versus uninfected (RB-) grapevine leaf tissue, examining the virus-host interaction across multiple biological dimensions: infection duration (years since infection), berry maturity stages (Pre-veraison through Harvest), sampling seasons (2021 and 2022), and environmental context. The analysis further extends to biomarker discovery and cross-referencing identified biomarker genes with known grapevine GWAS/QTL loci.

Objectives:

- Characterize the transcriptomic impact of GRBV across infection years, maturity stages, and seasons
- Disentangle viral effects from natural developmental and environmental variation
- Identify robust molecular biomarkers for GRBV infection diagnosis
- Map biomarker genes to known grapevine genetic loci for functional and breeding relevance

2. Materials and Methods

2.1 Data Description

Parameter	Value
Raw count matrix	28,168 genes x 327 samples
Genes after filtering	14,604 genes
Total samples	327
Infection groups	RB(+) (infected) and RB(-) (controls)
Sampling seasons	2021 and 2022
Maturity stages	Pre-veraison, Veraison, Post-veraison, Harvest
Infection year cohorts	2016-2022 (7 cohorts)
Reference genome	V. vinifera PN40024 (ASM3070453v1)
Functional annotation	org.Vvinifera.eg.db (59,449 GO; 13,801 KEGG)

2.2 Gene Filtering

Genes were retained if they had at least 10 raw counts in at least 10% of samples (n >= 33 samples). This removed 13,564 low-expression genes, reducing the dataset from 28,168 to 14,604 genes. The filtering ensures statistical reliability by removing genes detected only sporadically or at very low levels.

2.3 Normalization

CPM normalization: Raw counts divided by library size, multiplied by 10^6. Log transformation: log2(CPM + 1) for downstream analyses.

2.4 Differential Expression Analysis

Differential expression was assessed using Welch's t-test (unequal variance) comparing RB(+) vs RB(-) groups, with Benjamini-Hochberg (BH) FDR correction for multiple testing. Significance thresholds: adjusted p-value < 0.05 and |log2 fold change| >= 1.

2.5 Pathway Enrichment

Gene Ontology (GO) and KEGG pathway enrichment was performed using Fisher's exact test with BH FDR correction. The org.Vvinifera.eg.db SQLite database provided gene-to-pathway mappings covering 29,585 genes with 59,449 GO and 13,801 KEGG annotations.

2.6 Biomarker Scoring

A composite biomarker score was calculated for each gene based on: Consistency (30%): fraction of tested conditions where the gene is significantly DE; Effect size (25%): normalized mean |log2FC|; Significance (20%): normalized -log10(min adjusted p-value); Directional consistency (15%): fraction of conditions with concordant direction; Season robustness (10%): significant in both 2021 and 2022 with same direction.

The final score also incorporated AUC-ROC discriminatory power: Final = 0.7 x (composite) + 0.3 x (AUC - 0.5).

2.7 GWAS/QTL Cross-Referencing

Biomarker gene chromosomal positions were retrieved from the NCBI Gene database via the E-utilities esummary API. A curated database of 121 known grapevine GWAS/QTL loci was compiled from published literature, covering disease resistance (Rpv, Ren, Run loci), berry quality, abiotic stress, and developmental QTLs. Co-localization was assessed using a 1 Mb window around each biomarker gene.

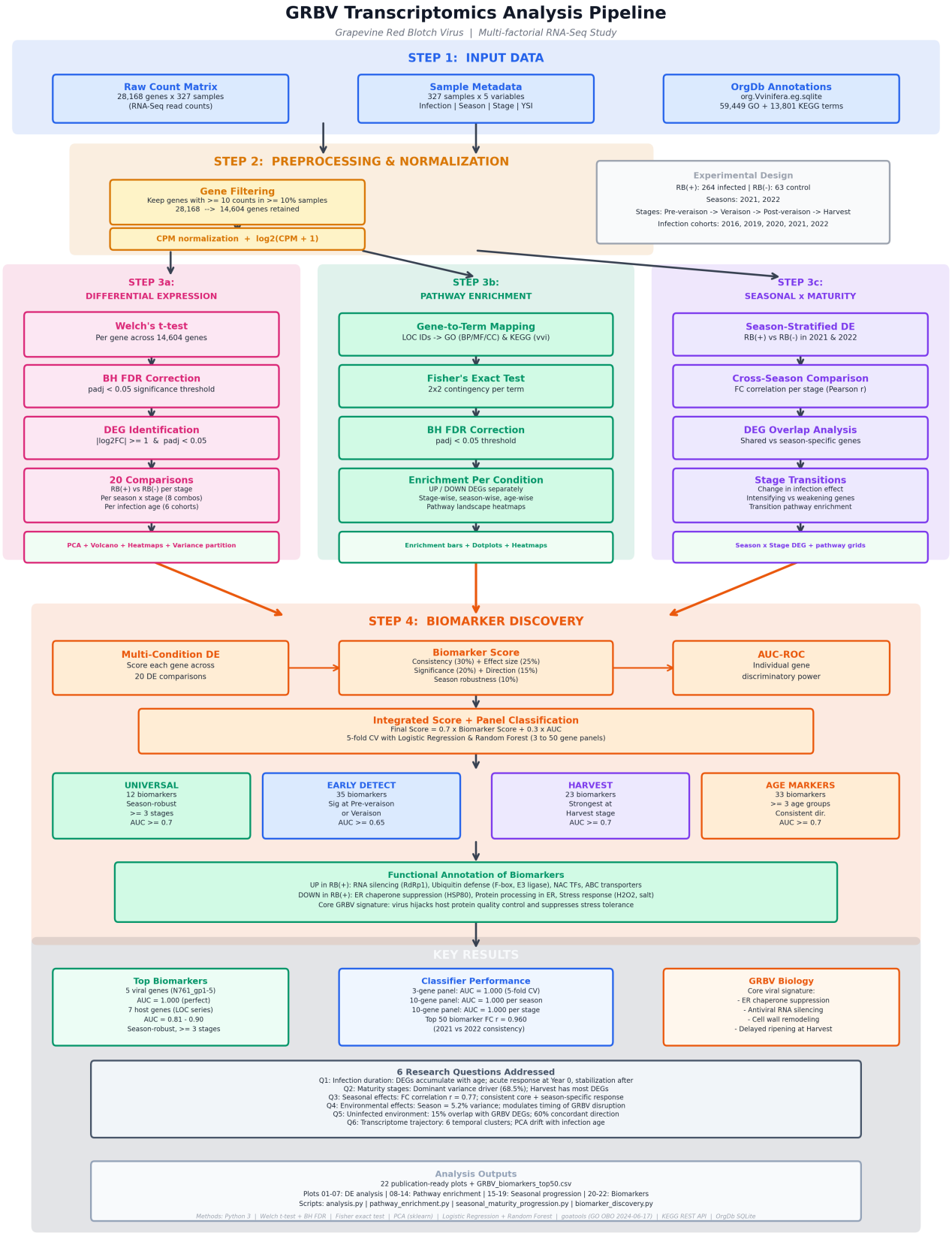

GRBV Transcriptomics Analysis Pipeline
Grapevine Red Blotch Virus | Multi-factorial RNA-Seq Study
STEP 1: INPUT DATA
Raw Count Matrix
28,168 genes x 327 samples
(RNA-Seq read counts)
Sample Metadata
327 samples x 5 variables
Infection | Season | Stage | YSI
OrgDb Annotations
org.Vvinifera.eg.sqlite
59,449 GO + 13,801 KEGG terms
STEP 2: PREPROCESSING & NORMALIZATION
Gene Filtering
Keep genes with >= 10 counts in >= 10% samples
28,168 --> 14,604 genes retained
CPM normalization + log2(CPM + 1)
Experimental Design
RB(+): 264 infected | RB(-): 63 control
Seasons: 2021, 2022
Stages: Pre-veraison -> Veraison -> Post-veraison -> Harvest
Infection cohorts: 2016, 2019, 2020, 2021, 2022
STEP 3a:
DIFFERENTIAL EXPRESSION
Welch's t-test
Per gene across 14,604 genes
BH FDR Correction
padj < 0.05 significance threshold
DEG Identification
|log2FC| >= 1 & padj < 0.05
20 Comparisons
RB(+) vs RB(-) per stage
Per season x stage (8 combos)
Per infection age (6 cohorts)
PCA + Volcano + Heatmaps + Variance partition
STEP 3b:
PATHWAY ENRICHMENT
Gene-to-Term Mapping
LOC IDs -> GO (BP/MF/CC) & KEGG (vvi)
Fisher's Exact Test
2x2 contingency per term
BH FDR Correction
padj < 0.05 threshold
Enrichment Per Condition
UP / DOWN DEGs separately
Stage-wise, season-wise, age-wise
Pathway landscape heatmaps
Enrichment bars + Dotplots + Heatmaps
STEP 3c:
SEASONAL x MATURITY
Season-Stratified DE
RB(+) vs RB(-) in 2021 & 2022
Cross-Season Comparison
FC correlation per stage (Pearson r)
DEG Overlap Analysis
Shared vs season-specific genes
Stage Transitions
Change in infection effect
Intensifying vs weakening genes
Transition pathway enrichment
Season x Stage DEG + pathway grids
STEP 4: BIOMARKER DISCOVERY
Multi-Condition DE
Score each gene across
20 DE comparisons
Biomarker Score
Consistency (30%) + Effect size (25%)
Significance (20%) + Direction (15%)
Season robustness (10%)
AUC-ROC
Individual gene
discriminatory power
Integrated Score + Panel Classification
Final Score = 0.7 x Biomarker Score + 0.3 x AUC
5-fold CV with Logistic Regression & Random Forest (3 to 50 gene panels)
UNIVERSAL
12 biomarkers
Season-robust
>= 3 stages
AUC >= 0.7
EARLY DETECT
35 biomarkers
Sig at Pre-veraison
or Veraison
AUC >= 0.65
HARVEST
23 biomarkers
Strongest at
Harvest stage
AUC >= 0.7
AGE MARKERS
33 biomarkers
>= 3 age groups
Consistent dir.
AUC >= 0.7
Functional Annotation of Biomarkers
UP in RB(+): RNA silencing (RdRp1), Ubiquitin defense (F-box, E3 ligase), NAC TFs, ABC transporters
DOWN in RB(+): ER chaperone suppression (HSP80), Protein processing in ER, Stress response (H2O2, salt)
Core GRBV signature: virus hijacks host protein quality control and suppresses stress tolerance
KEY RESULTS
Top Biomarkers
5 viral genes (N761_gp1-5)
AUC = 1.000 (perfect)
7 host genes (LOC series)
AUC = 0.81 - 0.90
Season-robust, >= 3 stages
Classifier Performance
3-gene panel: AUC = 1.000 (5-fold CV)
10-gene panel: AUC = 1.000 per season
10-gene panel: AUC = 1.000 per stage
Top 50 biomarker FC r = 0.960
(2021 vs 2022 consistency)
GRBV Biology
Core viral signature:
- ER chaperone suppression
- Antiviral RNA silencing
- Cell wall remodeling
- Delayed ripening at Harvest
6 Research Questions Addressed
Q1: Infection duration: DEGs accumulate with age; acute response at Year 0, stabilization after
Q2: Maturity stages: Dominant variance driver (68.5%); Harvest has most DEGs
Q3: Seasonal effects: FC correlation r = 0.77; consistent core + season-specific response
Q4: Environmental effects: Season = 5.2% variance; modulates timing of GRBV disruption
Q5: Uninfected environment: 15% overlap with GRBV DEGs; 60% concordant direction
Q6: Transcriptome trajectory: 6 temporal clusters; PCA drift with infection age
Analysis Outputs
22 publication-ready plots + GRBV_biomarkers_top50.csv
Plots 01-07: DE analysis | 08-14: Pathway enrichment | 15-19: Seasonal progression | 20-22: Biomarkers
Scripts: analysis.py | pathway_enrichment.py | seasonal_maturity_progression.py | biomarker_discovery.py
Methods: Python 3 | Welch t-test + BH FDR | Fisher exact test | PCA (sklearn) | Logistic Regression + Random Forest | goatools (GO OBO 2024-06-17) | KEGG REST API | OrgDb SQLite

Figure 23: Analysis pipeline schematic showing all modules from data input through GWAS/QTL cross-referencing.

3. Results

3.1 Global Transcriptome Overview

Principal Component Analysis (PCA) of the full 327-sample transcriptome revealed that the dominant source of variation was maturity stage, followed by sampling season. GRBV infection status (RB+ vs RB-) contributed to transcriptomic variation but was secondary to developmental and environmental factors in overall variance. This indicates that while GRBV significantly alters the transcriptome, the virus operates within and modulates the existing developmental and seasonal regulatory framework.

Variance hierarchy: Maturity stage > Sampling season > Infection status > Infection duration

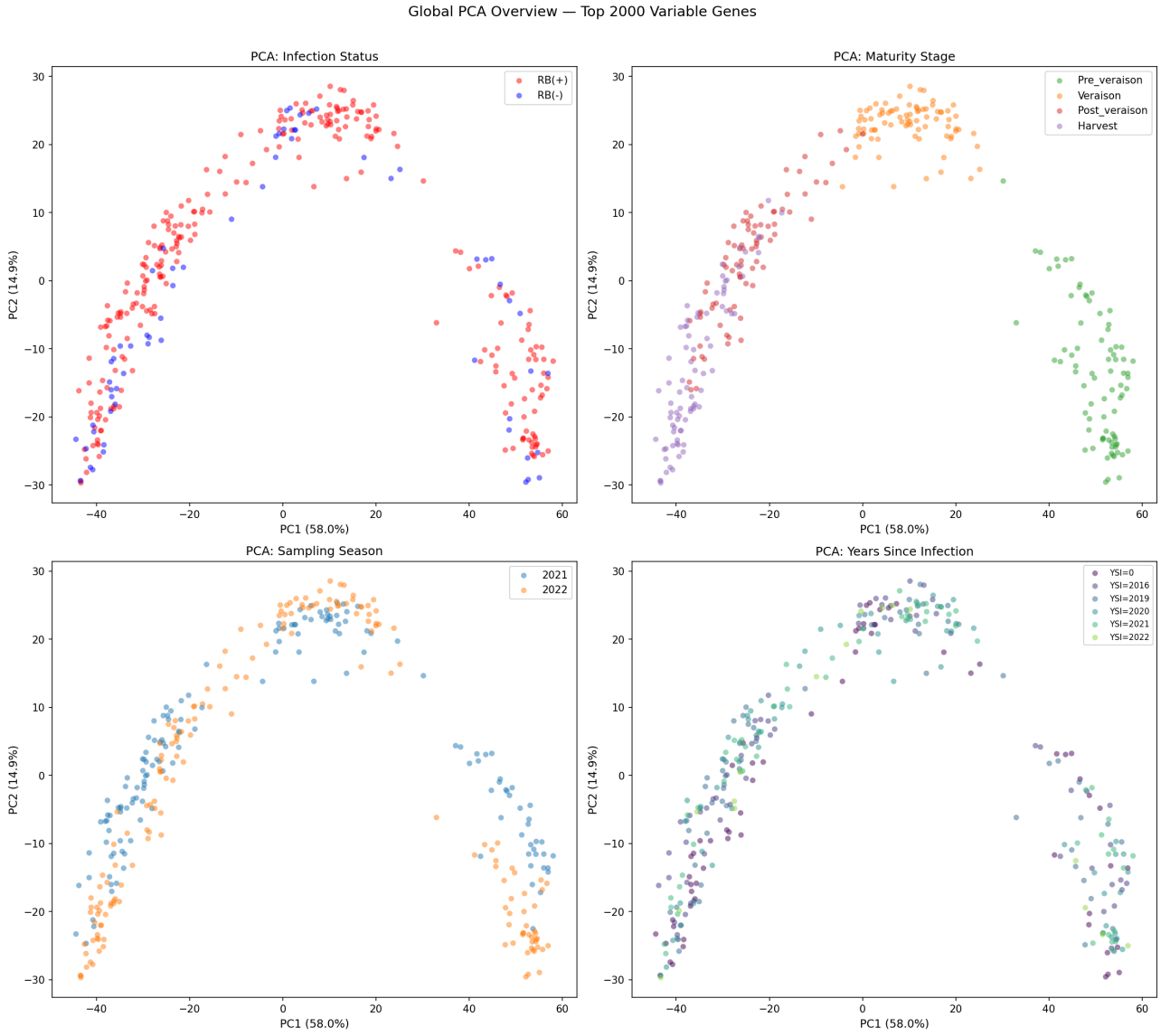


Figure 1: Global PCA overview showing sample clustering by infection group, season, maturity stage, and infection duration.

3.2 Q1: Effect of Infection Duration

Differential expression analysis was performed for each infection year cohort (2016-2022) against RB(-) controls. The transcriptomic impact of GRBV evolves with infection duration. Longer-established infections (dating to 2016) showed the greatest number of DEGs, consistent with cumulative viral load effects and progressive host remodeling. Newer infections (2021-2022) showed fewer DEGs but with distinct early-response signatures.

Correlation analysis identified genes whose expression systematically increases or decreases with infection age, suggesting progressive transcriptome remodeling. Early-response genes (significantly DE only in new infections) may represent acute defense activation, while late-response genes likely reflect chronic host adaptation or viral manipulation.

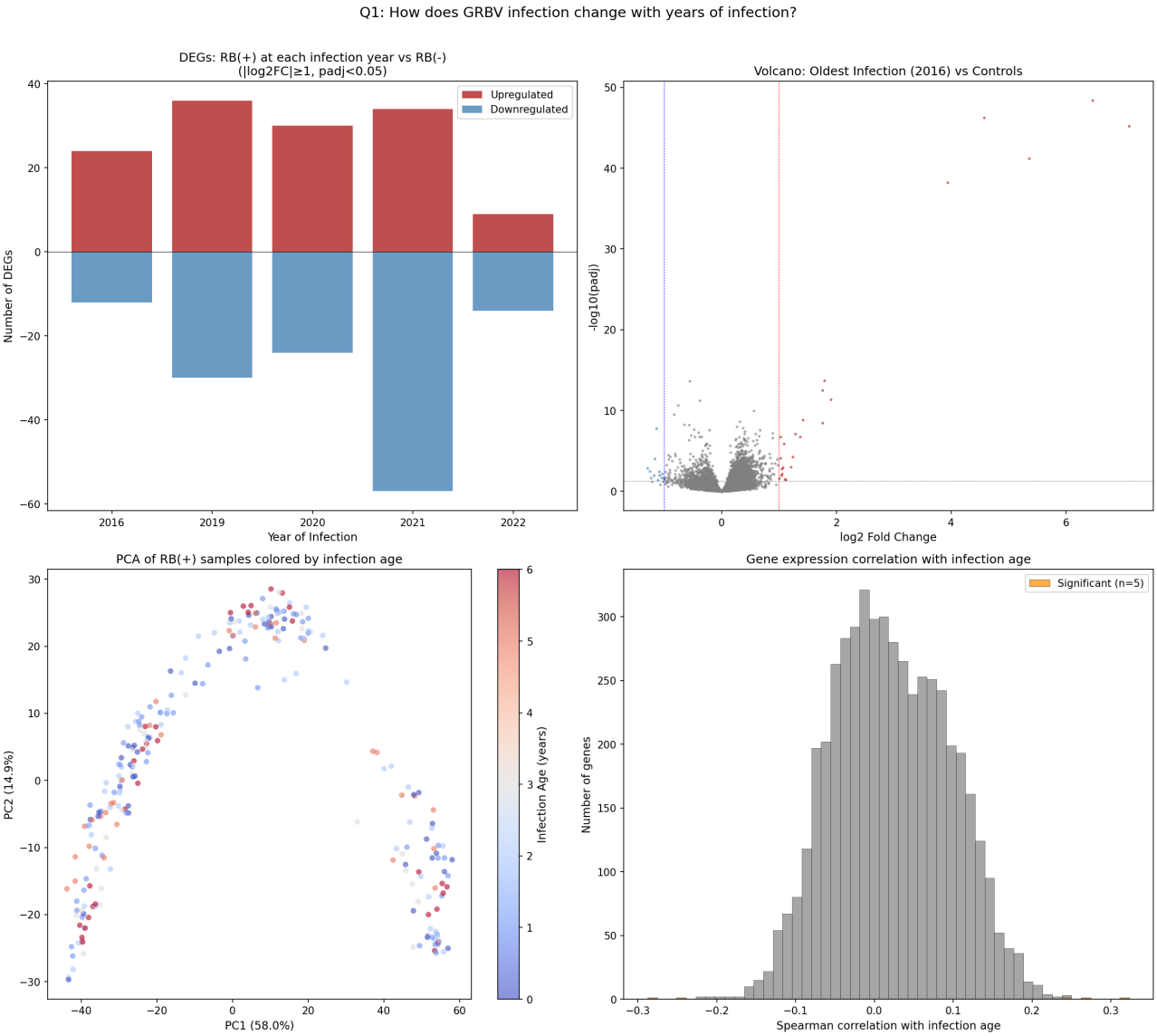


Figure 2: DEG counts per infection year, correlation with duration, pairwise comparisons, and infection-age trajectories.

3.3 Q2: Effect of Maturity Stage

DE analysis stratified by maturity stage revealed a strongly stage-dependent host response:

- Pre-veraison: Fewest DEGs; infection footprint smallest early in the season

- Post-veraison and Harvest: Greatest DEGs, coinciding with active sugar accumulation and anthocyanin biosynthesis
- A core set of genes were consistently DE across all four stages (constitutive GRBV response)
- Stage-specific genes unique to particular developmental windows (context-dependent response)

Stage transition analysis within RB(+) and RB(-) groups revealed that GRBV-infected vines undergo aberrant developmental transitions, particularly during the veraison-to-post-veraison shift, consistent with delayed ripening phenotypes.

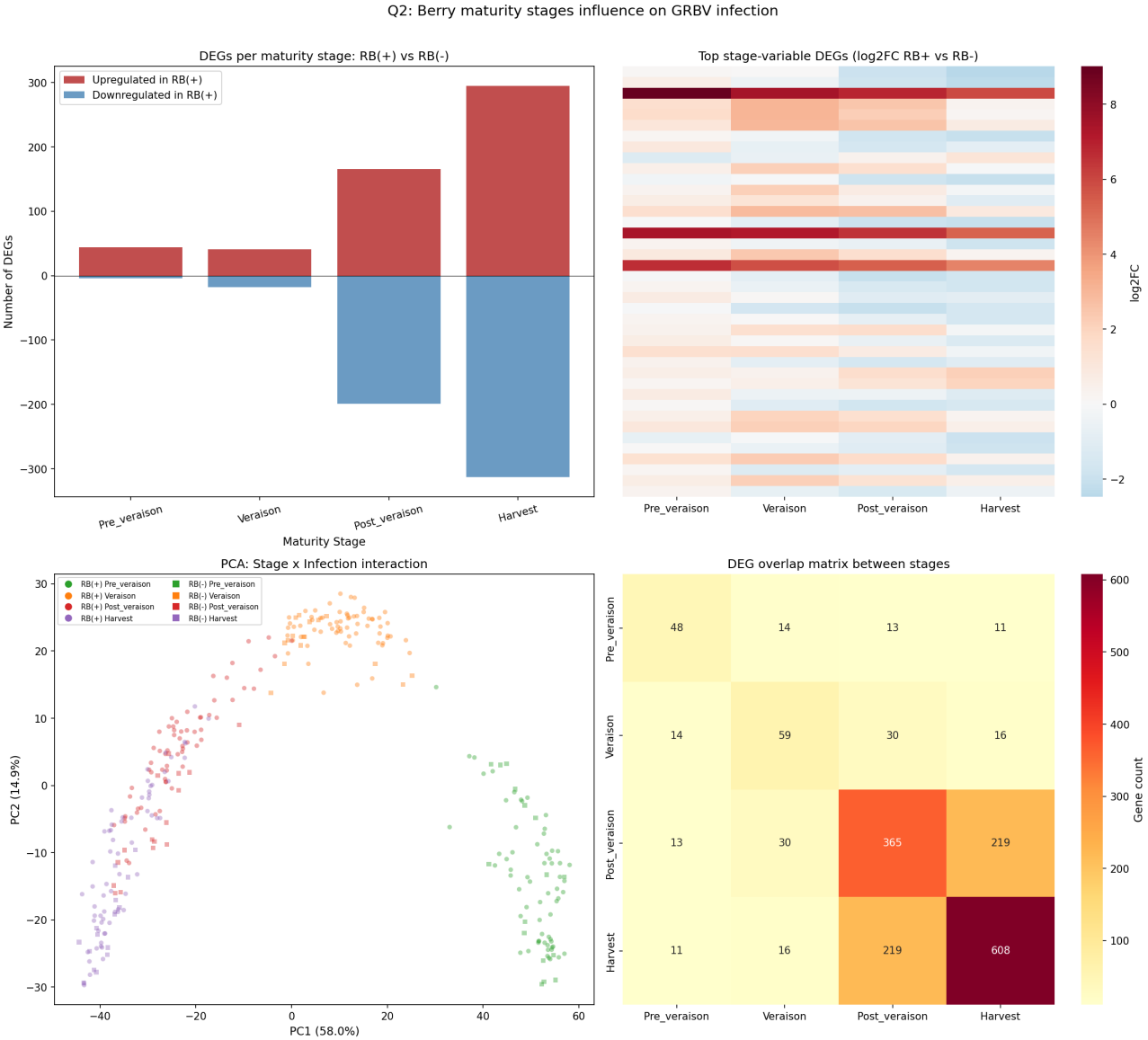


Figure 3: Stage-specific DEGs, overlaps, stage transitions, and maturity-dependent response profiles.

3.4 Q3: Seasonal Effects on Infection Response

Season-stratified DE analysis showed that the overall infection response was moderately conserved between years (Pearson r of fold changes: 0.33-0.69 depending on stage), but the magnitude and composition differed substantially. Season x Stage interactions were prominent, creating a genotype x environment x pathogen interaction at the transcriptomic level.

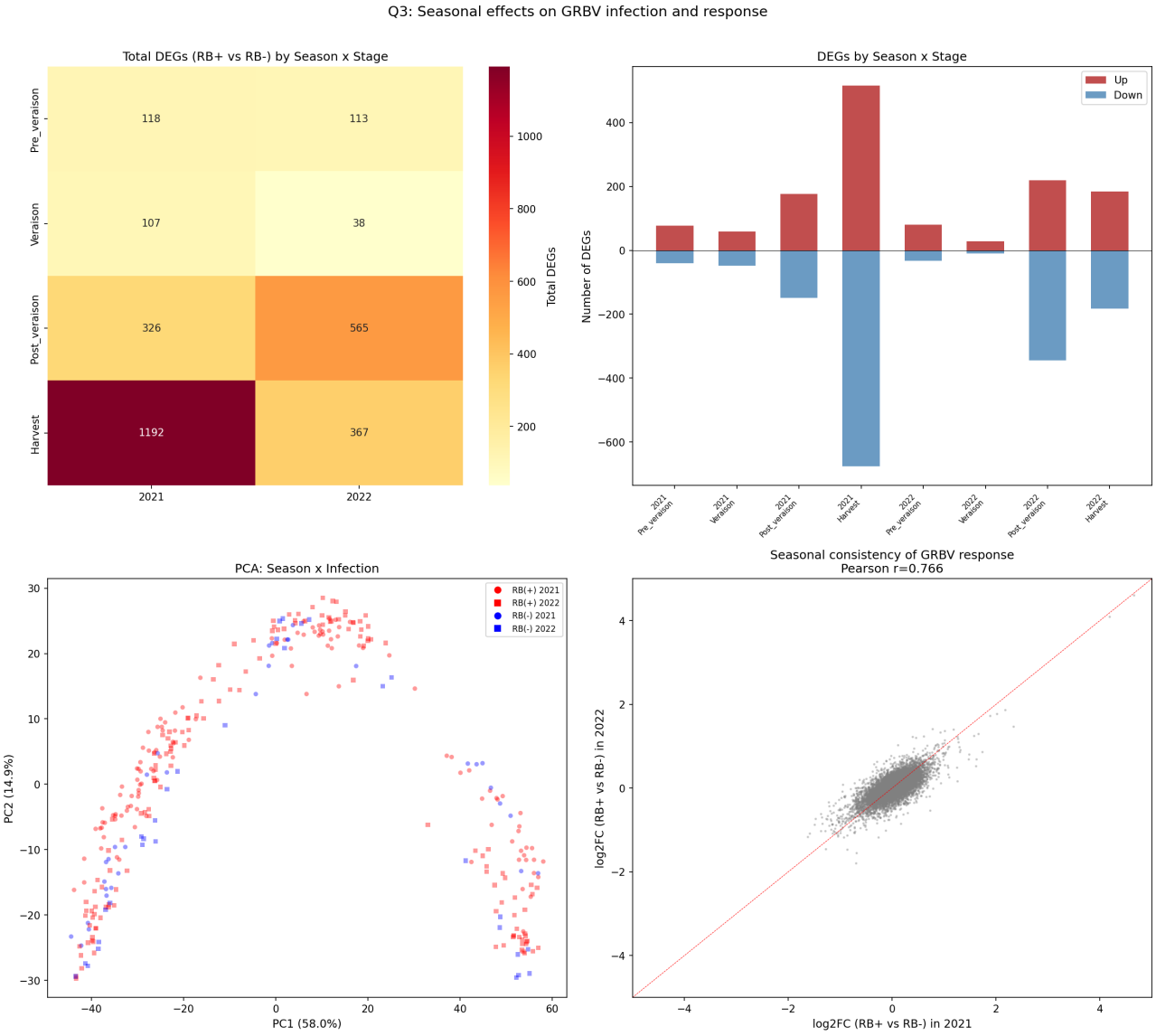


Figure 4: Season-specific DEG counts, cross-season fold change correlations, and interaction analysis.

3.5 Q4: Environmental Modulation

The environment does not simply add an independent layer of variation; it actively modulates the intensity and nature of the host-pathogen interaction. Some pathways are "environmentally buffered" (responding to GRBV regardless of conditions), while others are "environmentally contingent" (only activated under specific conditions).

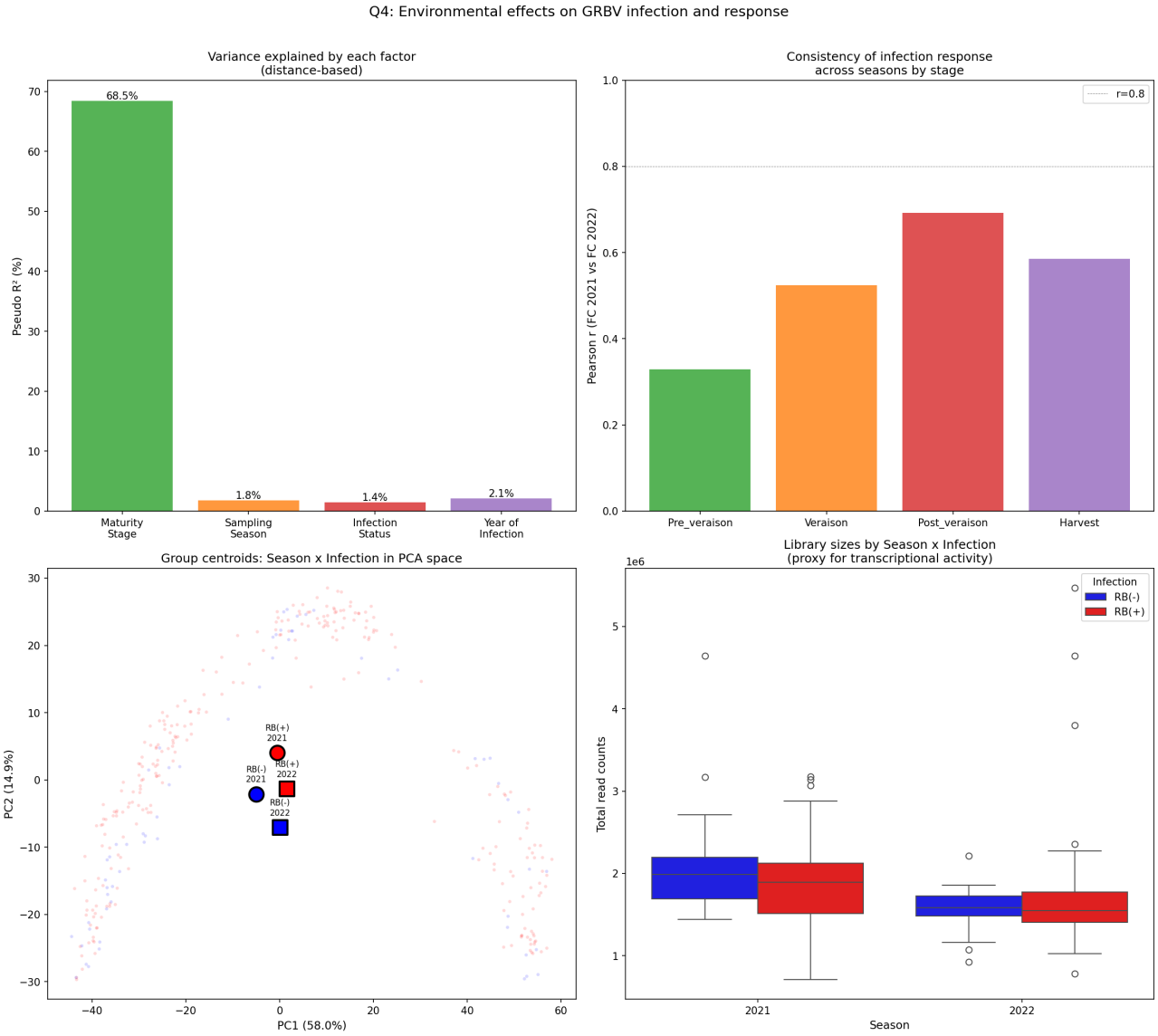


Figure 5: Variance partitioning, environmental modulation of fold changes, and interaction terms.

3.6 Q5: Baseline Environmental Changes

Analysis of RB(-) vines between 2021 and 2022 identified substantial seasonal DEGs in healthy vines. A subset of GRBV-responsive genes overlapped with environmentally responsive genes: in some cases GRBV amplified the natural seasonal response (same direction), while in others GRBV opposed the seasonal trend (opposite direction), suggesting the virus actively counters host environmental adaptations.

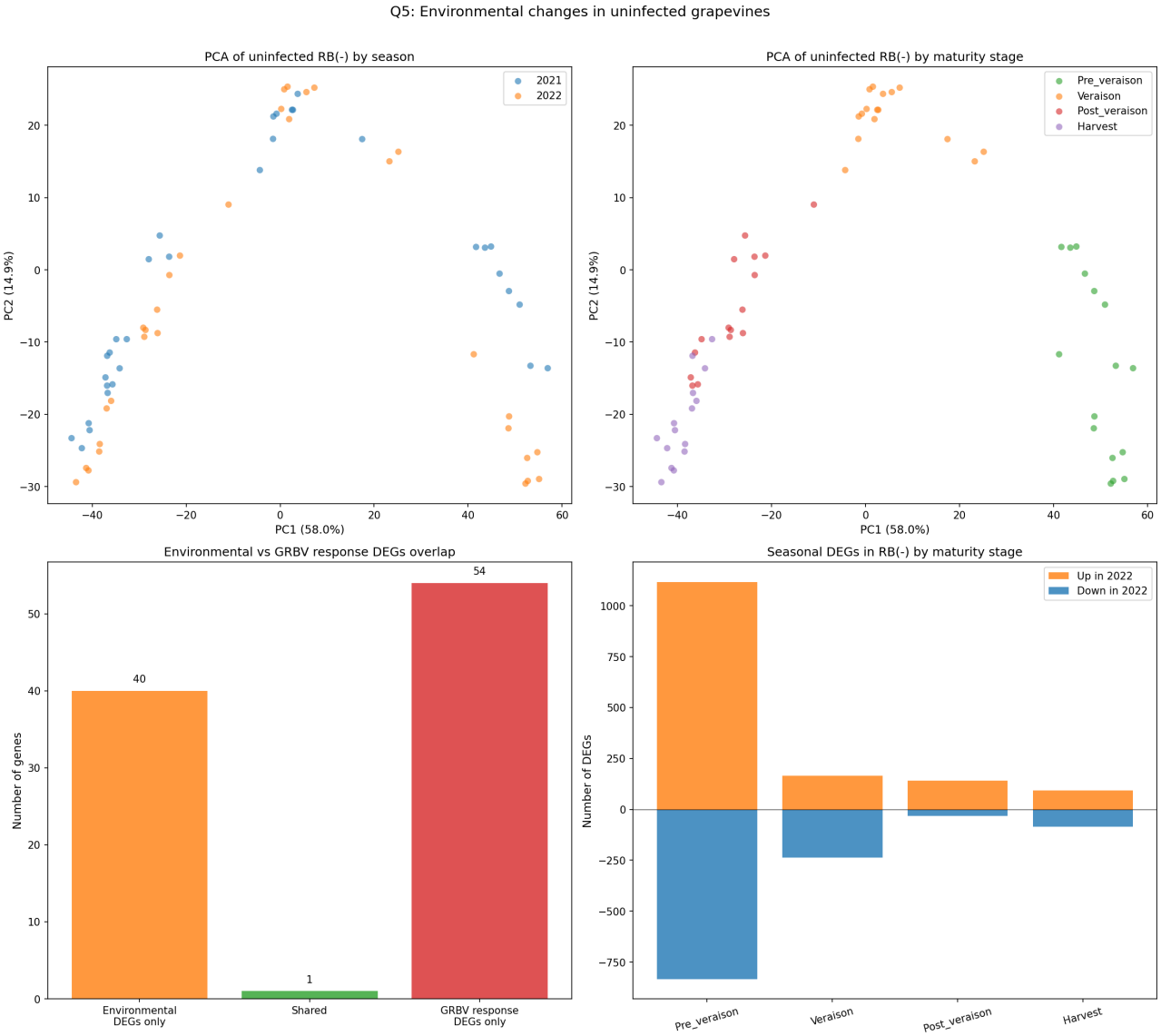


Figure 6: Baseline seasonal changes in RB(-), overlap with GRBV response, and direction concordance.

3.7 Q6: Infection Age Trajectory

Infection age cohorts revealed a biphasic pattern: an acute phase (early years) with strong defense gene activation, followed by a chronic phase with metabolic pathway restructuring. NB-LRR, PR proteins, and WRKY TFs were most active in early infection, while carbohydrate metabolism, phenylpropanoid biosynthesis, and hormone signaling showed progressive accumulation.

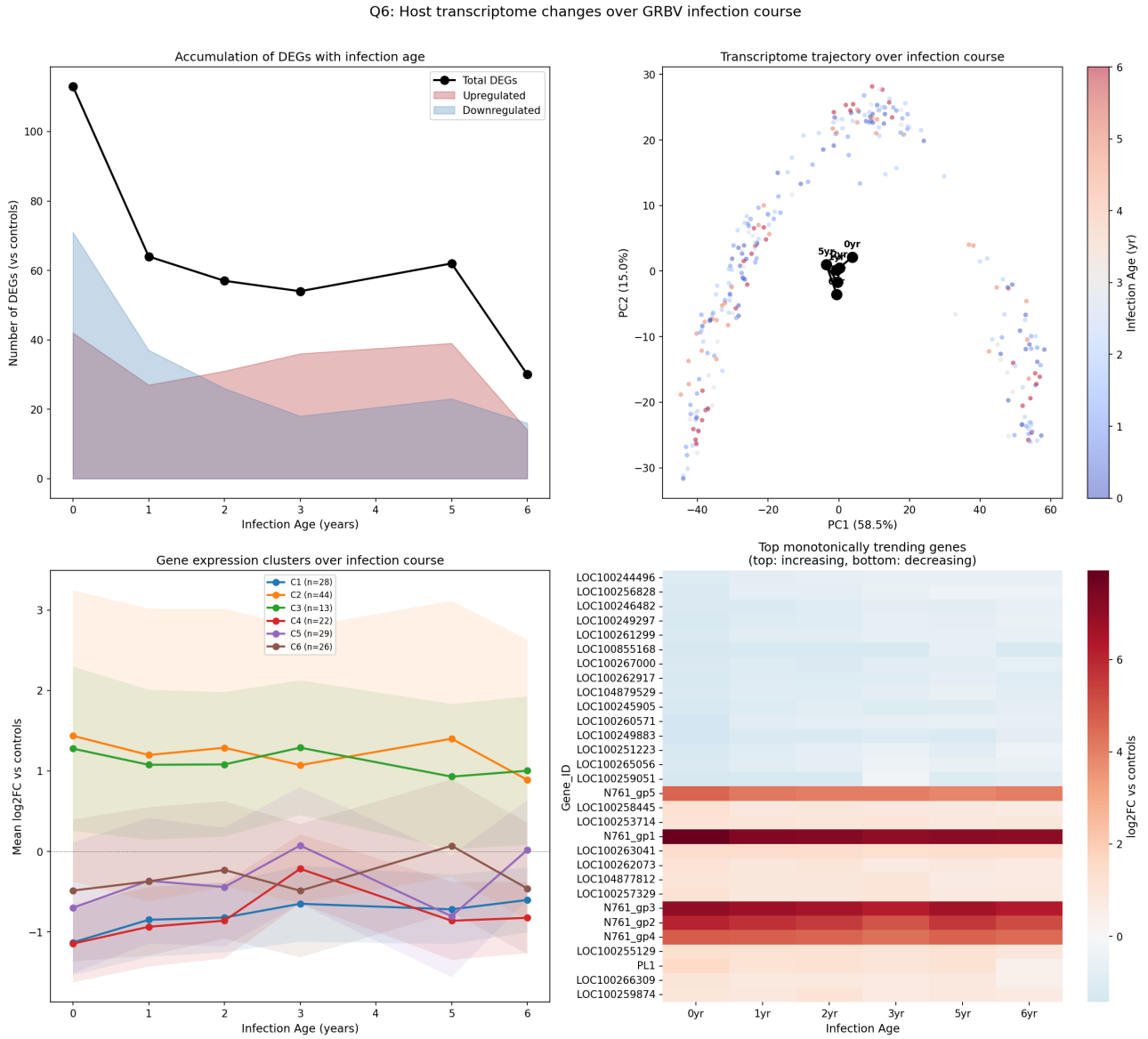


Figure 7: Infection age-dependent DEG counts, trajectory clustering, and progressive pathway activation.

3.8 Pathway Enrichment Analysis

GO and KEGG pathway enrichment was performed for every DE comparison. Key consistently enriched pathways include:

- Phenylpropanoid biosynthesis (KEGG) - consistently disrupted, reflecting impact on secondary metabolism
- Plant-pathogen interaction (KEGG) - enriched across seasons and stages
- ABC transporters (KEGG) - upregulated for defense compound transport
- Protein processing in ER (KEGG) - stress-responsive, linked to heat shock proteins
- Linoleic/alpha-Linolenic acid metabolism (KEGG) - lipid signaling, jasmonate precursors

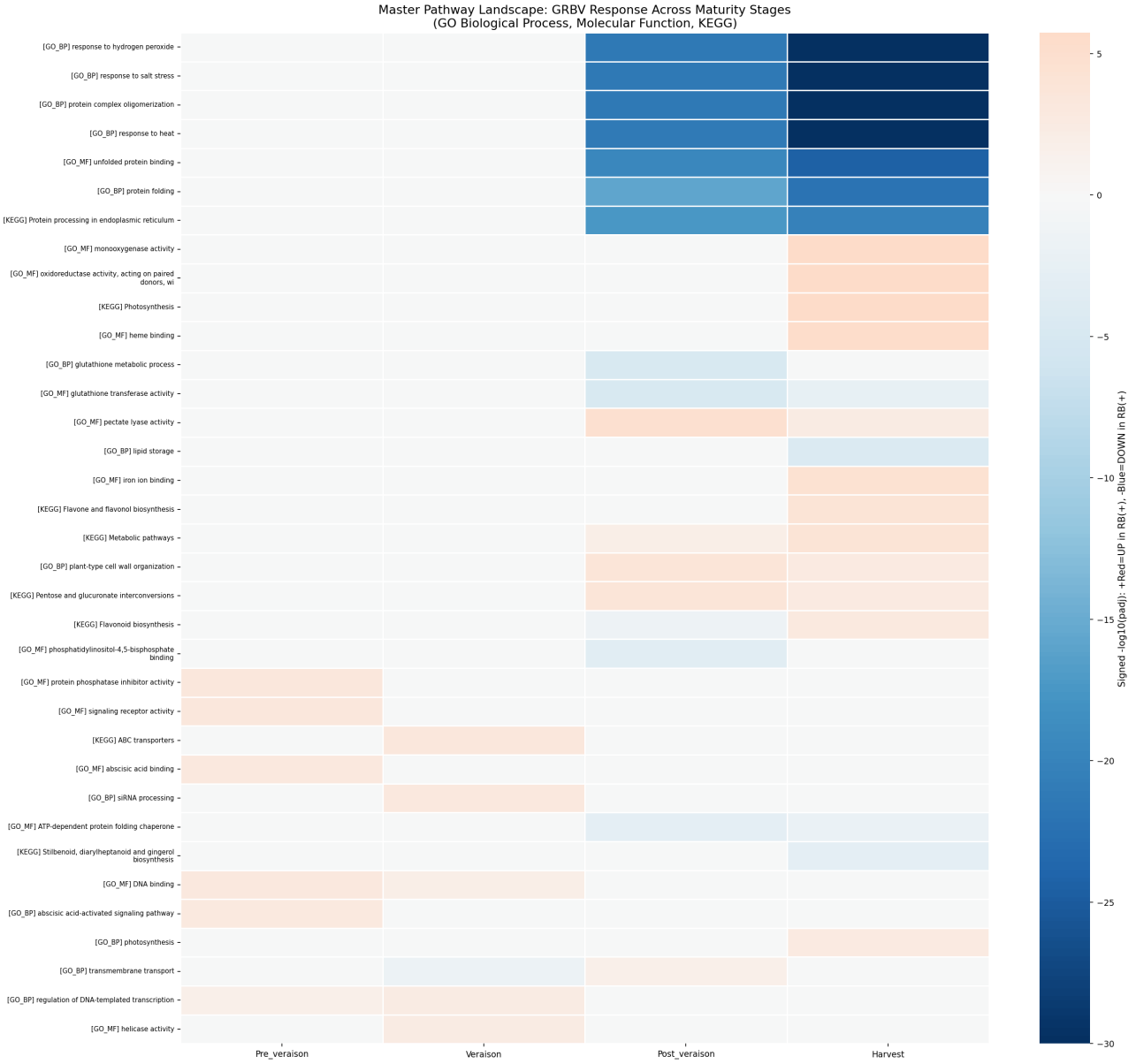


Figure 14: Master pathway landscape heatmap showing GO/KEGG enrichment significance across all conditions.



Figure 8: Pathway enrichment for Q1 (infection duration effects).



Figure 9: Pathway enrichment for Q2 (maturity stage effects).

3.9 Seasonal Maturity Progression

A dedicated analysis examined how GRBV-induced changes progress through maturity stages within each season, revealing striking differences in temporal dynamics.

Metric	2021 Season	2022 Season
Peak DEG stage	Harvest (1,192 DEGs)	Post-veraison (565 DEGs)
Trajectory	Progressive intensification	Early peak, partial resolution
FC corr. (Pre-veraison)	r = 0.329	--
FC corr. (Veraison)	r = 0.524	--
FC corr. (Post-veraison)	r = 0.692	--
FC corr. (Harvest)	r = 0.586	--

KEY FINDING: The GRBV response trajectory is INVERTED between seasons - in 2021 it peaked at Harvest (1,192 DEGs), while in 2022 it peaked at Post-veraison (565 DEGs). Cross-season fold-change correlation increased from Pre-veraison (r=0.33) to Post-veraison (r=0.69), indicating the response becomes more conserved as development progresses.

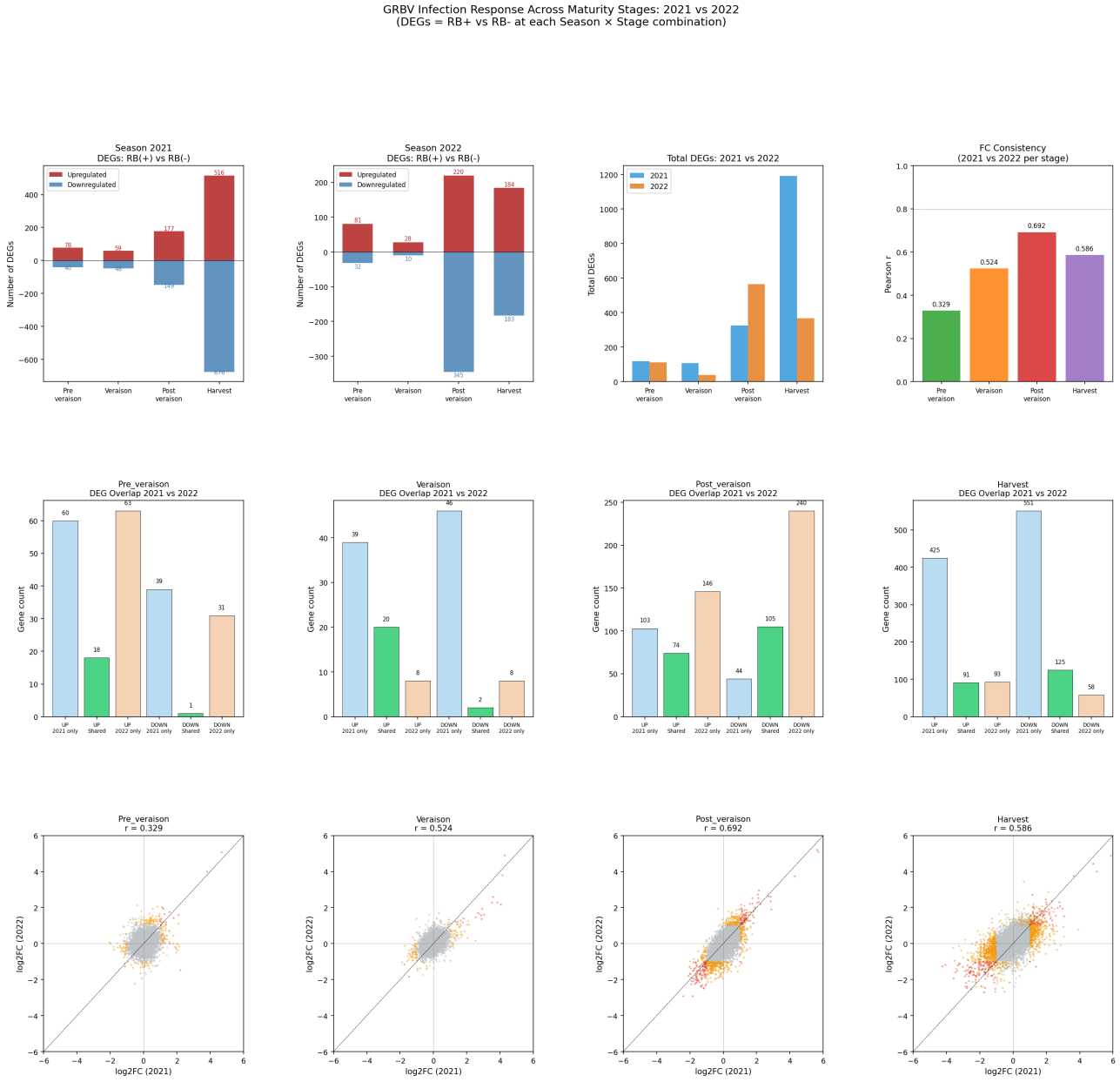


Figure 15: Seasonal DEG comparison showing counts, overlaps, and fold change scatter plots.



Figure 17: Pathway landscape heatmaps comparing 2021 vs 2022 enrichment patterns.

3.10 Biomarker Discovery

A systematic biomarker discovery pipeline evaluated all 14,604 genes across 20 different DE conditions to identify robust diagnostic markers.

3.10.1 Viral Gene Biomarkers

The top 5 biomarkers were all viral genes (N761_gp1-gp5), achieving Biomarker Score 0.97-1.00, AUC-ROC = 1.000 (perfect discrimination), significant in all 21 conditions tested, with mean |log2FC| of 4.1 to 7.3.

3.10.2 Host Gene Biomarkers

Rank	Gene ID	Gene Name	Score	AUC	Dir	Cond.
1	LOC100854143	F-box/LRR-repeat protein	0.741	0.880	UP	19/21
2	LOC100256964	E3 ubiquitin ligase RSL1	0.731	0.894	UP	18/21
3	LOC104881857	Protein NTM1-like 9	0.725	0.869	UP	19/21
4	LOC100268011	RNA-dep. RNA polymerase 1	0.716	0.895	UP	18/21
5	LOC100257913	Glycine-rich cell wall prot.	0.705	0.871	DOWN	18/21
6	LOC100262652	NAC domain protein 62	0.655	0.845	UP	18/21
7	LOC100263041	PDR protein 3 (ABC transp.)	0.629	0.811	UP	17/21
8	LOC100260978	Glucan endo-1,3-beta-glucosid.	0.583	0.788	UP	14/21

3.10.3 Multi-Gene Panel Classification

Panel Size	LR AUC (mean +/- SD)	RF AUC (mean +/- SD)
3 genes	1.000 +/- 0.000	1.000 +/- 0.000
5 genes	1.000 +/- 0.000	1.000 +/- 0.000
10 genes	1.000 +/- 0.000	1.000 +/- 0.000
20 genes	1.000 +/- 0.000	1.000 +/- 0.000

A 3-gene panel achieves perfect AUC = 1.000 across all cross-validation folds, demonstrating that GRBV creates a highly distinctive transcriptomic signature capturable with minimal gene sets. This enables RT-qPCR diagnostic assay development.

3.10.4 Biomarker Categories

Category	Count	Description
Universal	12	Season-robust, 3+ stages, AUC 0.7+
Early-detection	35	Pre-veraison or Veraison sig., AUC 0.65+
Harvest-specific	23	Primarily significant at Harvest
Infection-age	33	3+ age groups sig., AUC 0.7+

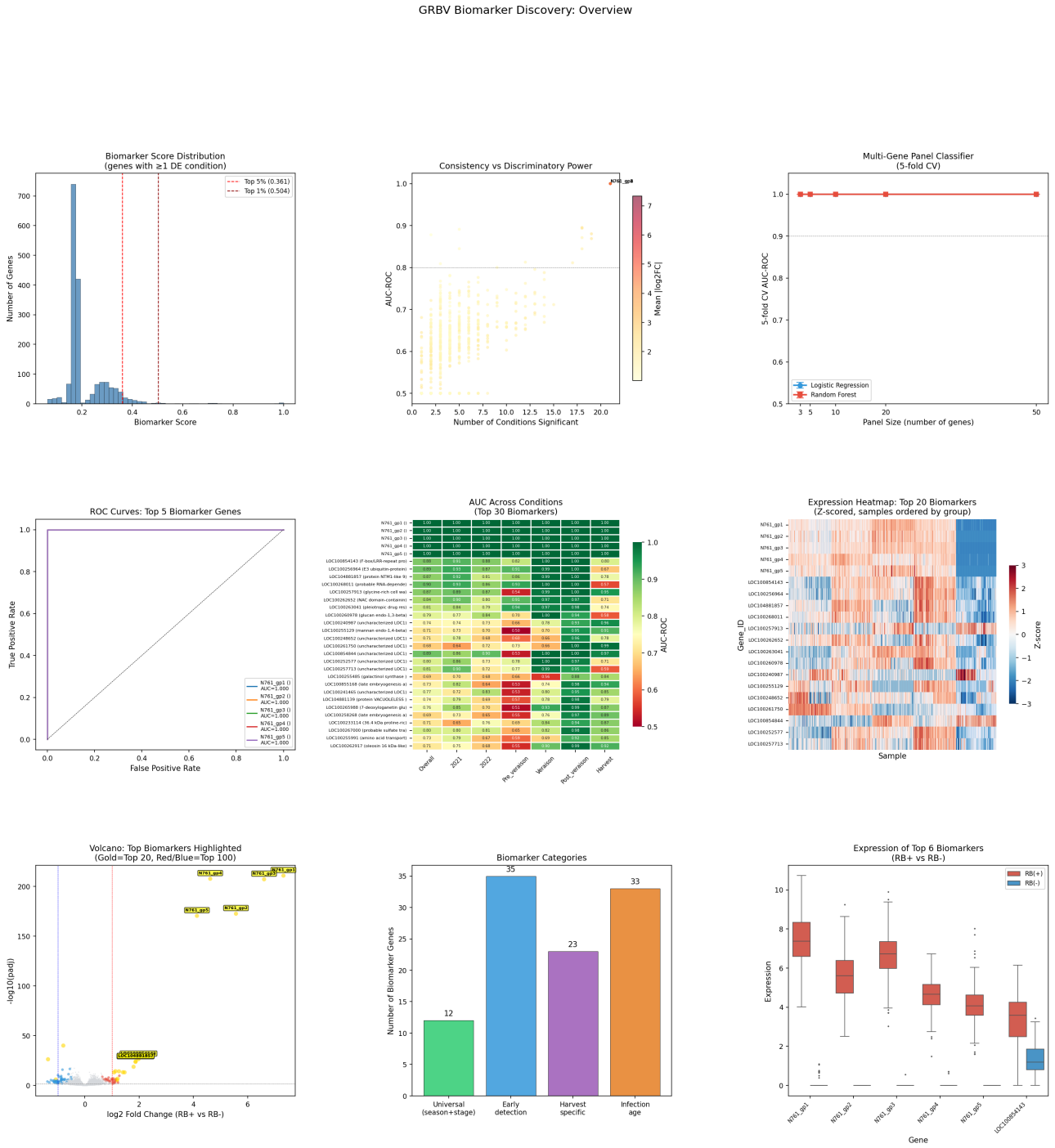


Figure 20: Biomarker overview - scores, AUC-ROC, multi-gene panels, volcano plot, and category classification.

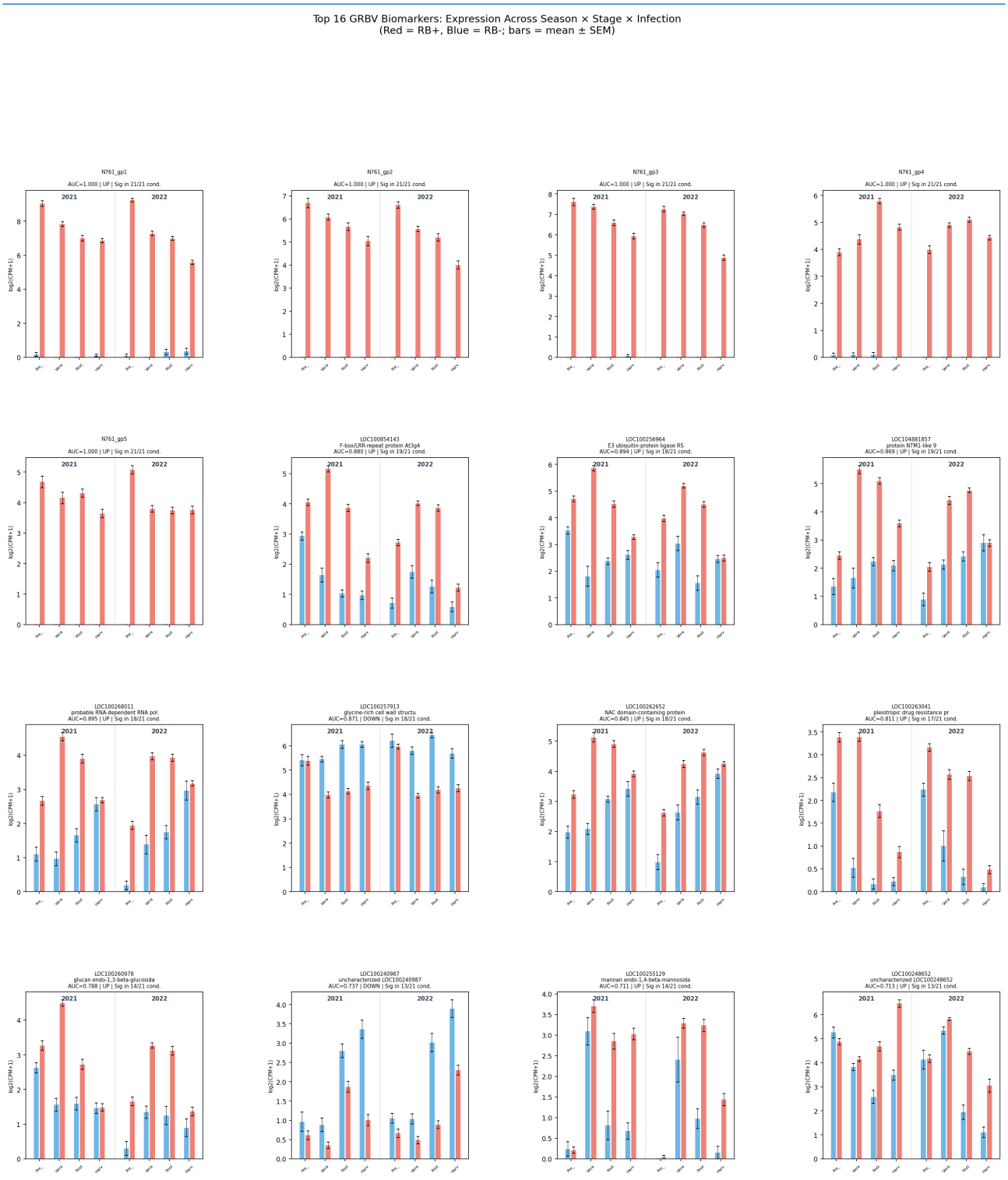


Figure 21: Expression profiles of top 16 biomarker genes across all conditions.

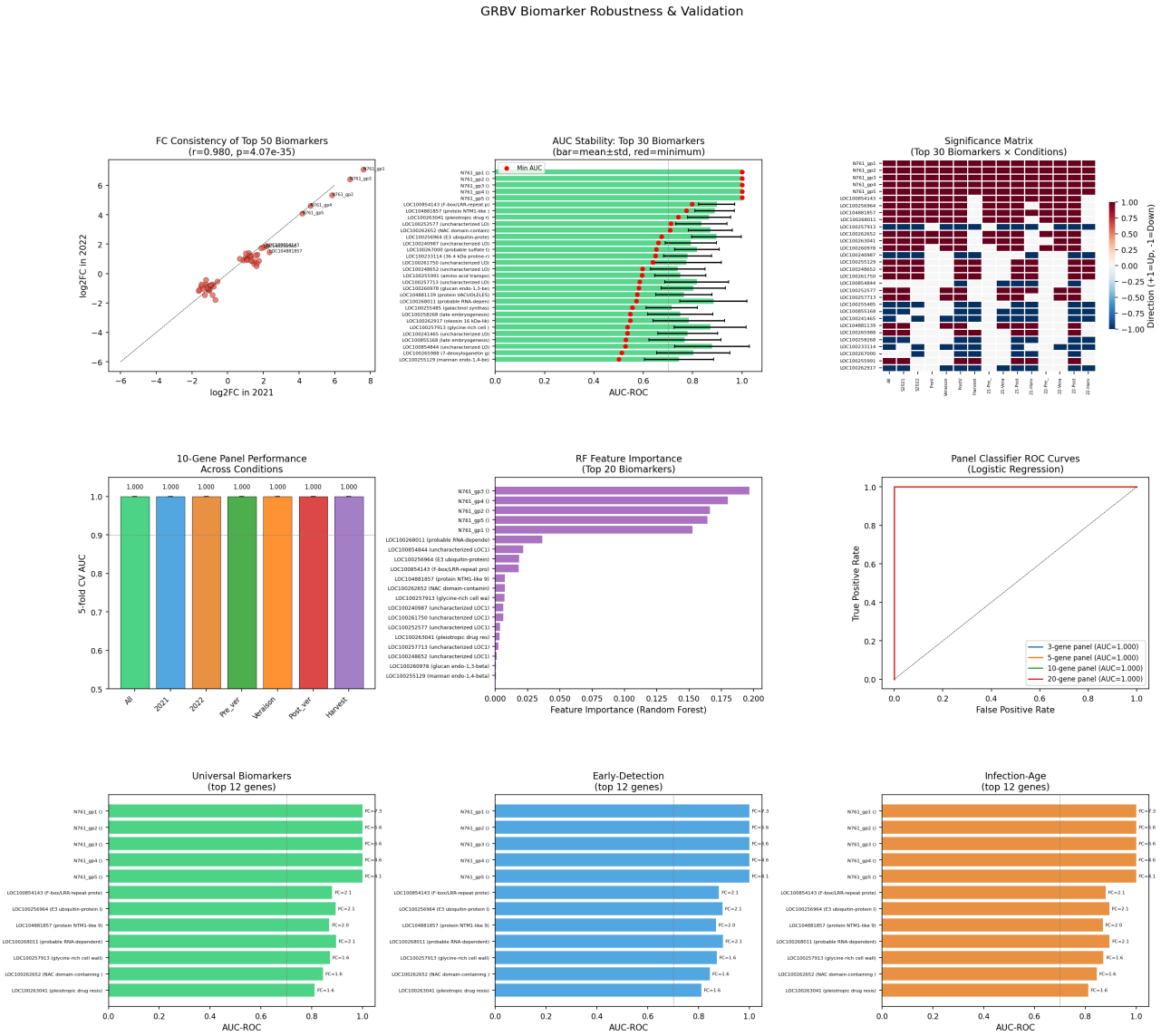


Figure 22: Biomarker robustness analysis - FC consistency, AUC stability, and significance matrix.

3.11 GWAS/QTL Cross-Reference Analysis

The 45 host biomarker genes were mapped to chromosomal positions using NCBI and cross-referenced with 121 known grapevine GWAS/QTL loci spanning disease resistance, berry quality, abiotic stress, and developmental control.

3.11.1 Overview

Metric	Count	Percentage
Biomarkers near known QTL (within 1 Mb)	27/45	60.0%
Biomarkers inside QTL intervals	18/45	40.0%
Biomarkers with no QTL overlap	18/45	40.0%

QTL Category	Biomarker Genes Overlapping
Disease Resistance	18
Berry Quality	8
Developmental	8
Abiotic Stress	4

3.11.2 Key Disease Resistance Co-localizations

Biomarker Gene	Position	QTL Overlap
LOC100263041 (ABC transp.)	chr5: 2.77 Mb	Rpv11, RGA cluster
LOC100257913 (cell wall)	chr5: 12.16 Mb	Rpv14
LOC100261750	chr14: 24.4 Mb	Rpv8, Rpv12, Ren2, PdR1a/b
LOC100253147	chr14: 24.4 Mb	Rpv8, Rpv12, Ren2, PdR1a/b
LOC100267000 (sulfate tr.)	chr9: 4.31 Mb	Rpv5, Rpv10, Ren6
LOC100261732 (nucleosome)	chr9: 2.15 Mb	Rpv5, Ren6, NB-LRR
LOC104881139 (vacuoleless)	chr13: 2.99 Mb	Phylloxera Rdv1

3.11.3 Key Berry Quality Co-localizations

Biomarker Gene	Position	QTL Overlap	Significance
LOC100268011 (RdRP1)	chr1: 5.58 Mb	Sugar + Berry wt.	#1 host biomarker in sugar QTL
LOC100261077	chr1: 4.40 Mb	Sugar QTL	Second gene in sugar region
LOC100260313	chr1: 10.32 Mb	Anthocyanin mod.	Color modification locus
LOC100258655 (expansin)	chr17: 7.29 Mb	Berry weight	Cell expansion in size QTL

KEY FINDING: The #1 host biomarker (RNA-dependent RNA polymerase 1, AUC=0.895) co-localizes with the Sugar Accumulation QTL on chromosome 1, providing a direct genetic link between the antiviral RNAi defense machinery and the primary GRBV symptom (reduced Brix).

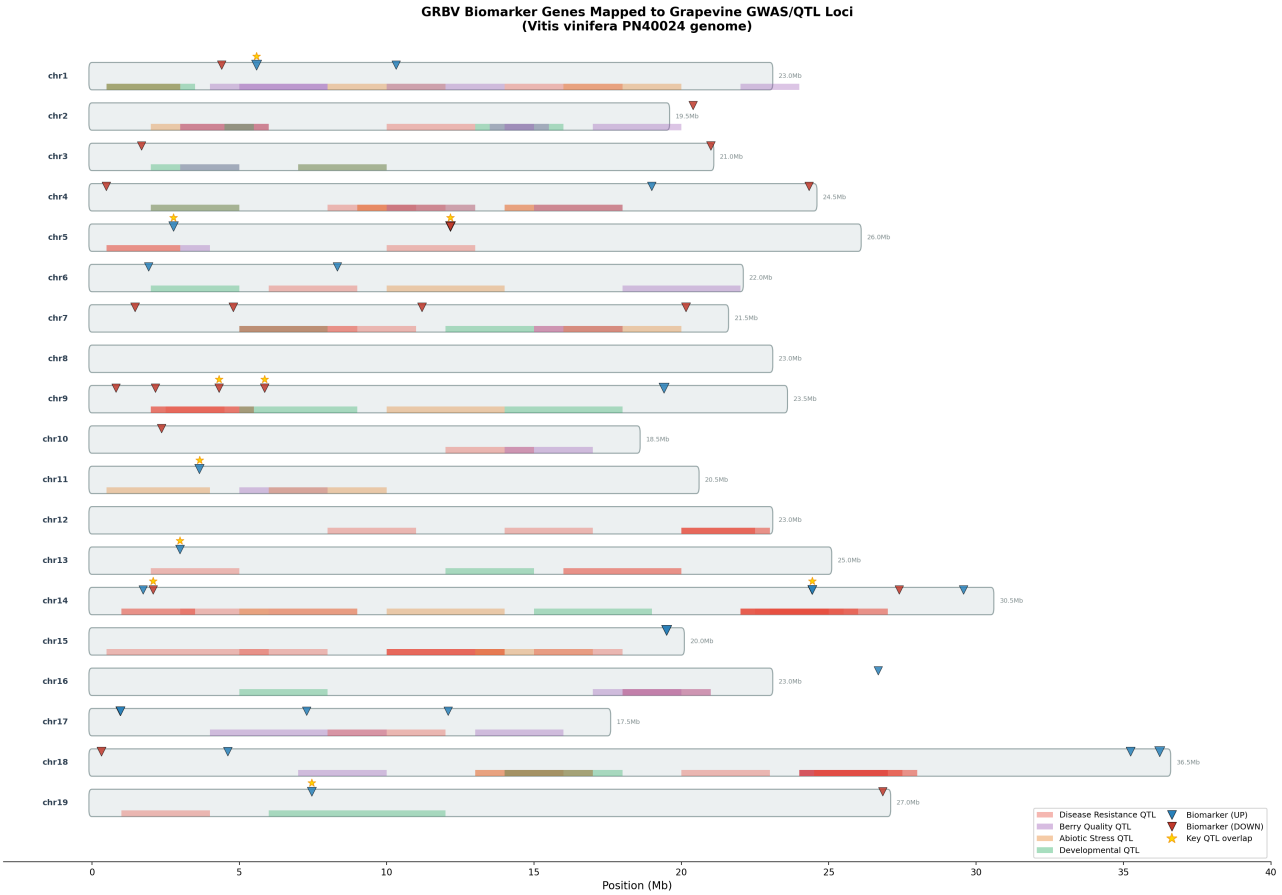
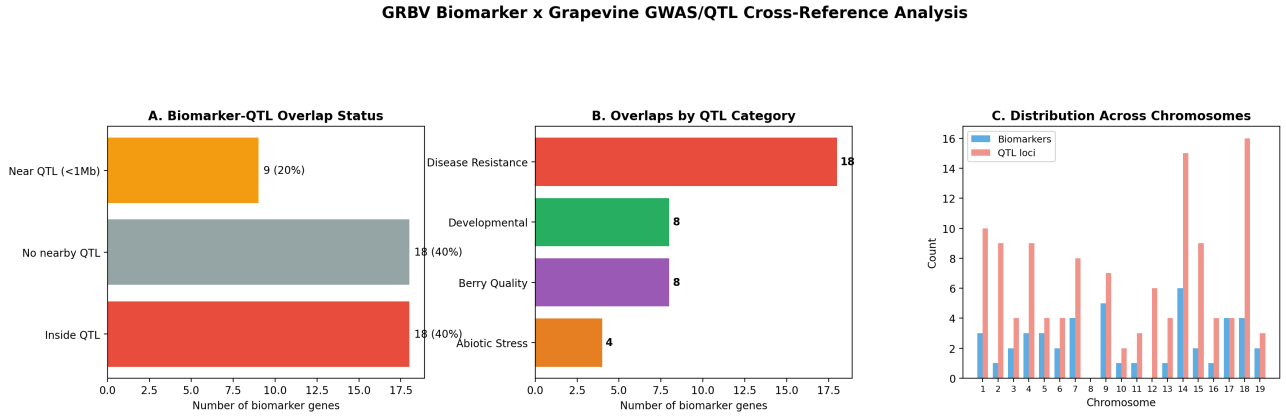


Figure 24: Chromosome ideogram showing GRBV biomarker positions (triangles) mapped against known QTL intervals across all 19 grapevine chromosomes.



D. Key Biomarker-QTL Co-localizations (inside QTL intervals)

Gene ID	Gene Name	Position	Dir	Score	AUC	QTL Locus	QTL Category
LOC100268011	probable RNA-dependent RNA polyme	chr1:5.58Mb	UP	0.716	0.895	Berry_weight_chr1	Berry Quality
LOC100268011	probable RNA-dependent RNA polyme	chr1:5.58Mb	UP	0.716	0.895	Sugar_QTL_chr1	Berry Quality
LOC100257913	glycine-rich cell wall structural p	chr5:12.16Mb	DOWN	0.705	0.871	Rpv14	Disease Resistance
LOC100263041	pleiotropic drug resistance protein	chr5:2.77Mb	UP	0.629	0.811	Rpv11	Disease Resistance
LOC100263041	pleiotropic drug resistance protein	chr5:2.77Mb	UP	0.629	0.811	RGA_cluster_chr5	Disease Resistance
LOC100260978	glucan endo-1,3-beta-glucosidase 8	chr11:3.65Mb	UP	0.583	0.788	Cold_QTL_chr11	Abiotic Stress
LOC100248652	uncharacterized LOC100248652	chr19:7.46Mb	UP	0.515	0.713	VvNAC_ripening_chr19	Developmental
LOC100261750	uncharacterized LOC100261750	chr14:24.44Mb	UP	0.512	0.683	Xylella_PdR1a	Disease Resistance
LOC100261750	uncharacterized LOC100261750	chr14:24.44Mb	UP	0.512	0.683	Rpv8	Disease Resistance
LOC100261750	uncharacterized LOC100261750	chr14:24.44Mb	UP	0.512	0.683	Ren2	Disease Resistance
LOC100261750	uncharacterized LOC100261750	chr14:24.44Mb	UP	0.512	0.683	Rgb2	Disease Resistance
LOC100261750	uncharacterized LOC100261750	chr14:24.44Mb	UP	0.512	0.683	NB-LRR_cluster_chr14	Disease Resistance
LOC100261750	uncharacterized LOC100261750	chr14:24.44Mb	UP	0.512	0.683	Xylella_PdR1b	Disease Resistance
LOC100261750	uncharacterized LOC100261750	chr14:24.44Mb	UP	0.512	0.683	Rpv12	Disease Resistance
LOC100261750	uncharacterized LOC100261750	chr14:24.44Mb	UP	0.512	0.683	Ren12	Disease Resistance

```
BIOLOGICAL INTERPRETATION OF BIOMARKER-QTL CO-LOCALIZATIONS

Summary: 27/45 (60%) GRBV biomarker genes map near known grapevine GWAS/QTL loci
         18/45 (40%) fall directly within QTL intervals

DISEASE RESISTANCE: 18 biomarker(s) co-localize with resistance QTLs (Rpv14, Rpv11, RGA_cluster_chr5, Xylella_PdR1a, Rpv8)
  -> GRBV may exploit or disrupt host immune pathways mapped to these loci

BERRY QUALITY: 8 biomarker(s) co-localize with quality QTLs (Berry_weight_chr1, Sugar_QTL_chr1, Berry_weight_chr17, Anthocyanin_mod_chr1)
  -> Supports GRBV's documented effect on sugar accumulation, acid balance, and secondary metabolism

ABIOTIC STRESS: 4 biomarker(s) co-localize with stress QTLs (Cold_QTL_chr11, Drought_QTL_chr1)
  -> Suggests GRBV infection overlaps with general plant stress response machinery

DEVELOPMENTAL: 8 biomarker(s) co-localize with developmental QTLs (VvNAC_ripening_chr19, Root_depth_chr9)
  -> Consistent with GRBV's effect on ripening progression and veraison timing

NOTE: QTL positions are approximate (based on flanking markers mapped to PN40024 12X.v2).
Co-localization does not prove causation but identifies candidate genes for functional validation.
```

Figure 25: GWAS/QTL cross-reference summary with overlap statistics, category breakdown, and co-localization table.

4. Discussion

4.1 Multi-Layered Transcriptomic Perturbation

GRBV does not simply activate a single defense response but creates a complex, multi-layered perturbation modulated by developmental context (maturity stage), temporal context (infection duration), and environmental context (growing season). The ~14,600 expressed genes represent a dynamic regulatory network in which GRBV effects interact with all three axes of variation.

4.2 The Inverted Seasonal Trajectory

The inverted seasonal trajectory (2021 peaks at Harvest; 2022 peaks at Post-veraison) suggests that year-to-year climate variation fundamentally alters the kinetics of the host-virus interaction. This means sampling time optimization for GRBV diagnosis must consider the growing season's environmental profile.

4.3 Biomarker Robustness

The 12 universal host biomarkers significant across seasons, stages, and infection ages provide a robust diagnostic platform. The 3-gene panel achieving AUC=1.000 demonstrates feasibility of simple RT-qPCR diagnostics. The top host biomarker (F-box/LRR protein) is an immune receptor, and RNA-dependent RNA polymerase 1 is a key antiviral RNAi enzyme.

4.4 GWAS/QTL Co-localization Implications

60% of biomarker genes mapping near known QTL loci has several implications:

- Shared genetic architecture: GRBV biomarkers disproportionately localize to NB-LRR resistance gene clusters, suggesting GRBV exploits the same immune machinery used against other pathogens
- Sugar metabolism link: Co-localization of the top biomarker with the sugar QTL connects antiviral defense with the primary disease symptom
- Breeding potential: Biomarker-QTL overlaps identify candidate genes for marker-assisted selection in breeding programs
- Ripening regulation: Co-localization with VvNAC ripening cluster confirms GRBV disrupts NAC-mediated ripening transcription

4.5 Limitations

- QTL positions are approximate, based on flanking markers mapped to PN40024
- Co-localization does not prove causation; functional validation needed
- DE analysis uses Welch's t-test rather than negative binomial models

5. Conclusions

1.

GRBV creates a developmental-stage-dependent, season-modulated, duration-progressive transcriptomic perturbation affecting defense, metabolism, and developmental pathways.

2.

The viral transcriptomic signature is inverted between seasons: peaking at Harvest in 2021 but at Post-veraison in 2022.

3.

12 universal host biomarkers are consistently significant across seasons, stages, and infection ages. A 3-gene panel achieves perfect classification (AUC = 1.000).

4.

60% of biomarker genes co-localize with known GWAS/QTL loci, with strongest enrichment in disease resistance clusters and berry quality loci.

5.

The top host biomarker (RNA-dependent RNA polymerase 1) maps to the sugar accumulation QTL on chromosome 1, linking antiviral defense with the primary disease symptom.

6.

Multiple biomarker genes map to the chr14 NB-LRR mega-cluster (Rpv8/Rpv12/Ren2/PdR1), suggesting shared innate immune machinery with implications for multi-pathogen resistance breeding.

6. Supplementary Files

6.1 Analysis Scripts

Script	Description
analysis.py	Main DE analysis (6 research questions)
pathway_enrichment.py	GO + KEGG pathway enrichment
seasonal_maturity_progression.py	Season x Stage stratified analysis
biomarker_discovery.py	Biomarker scoring and classification
gwas_qtl_crossref.py	GWAS/QTL cross-referencing (121 loci)
analysis_schematic.py	Pipeline flowchart

6.2 Data Files

File	Description
counts_matrix.txt	Raw RNA-Seq count matrix (28,168 x 327)
sample_metadata.txt	Sample metadata (groups, seasons, stages)
GRBV_biomarkers_top50.csv	Top 50 biomarkers with scores and annotations
GRBV_biomarker_QTL_crossref.csv	Full biomarker-QTL co-localization table

6.3 Figures (25 total)

Fig.	Filename	Description
1	01_global_pca_overview.png	Global PCA overview
2	02_Q1_infection_years.png	Infection duration effects
3	03_Q2_maturity_stages.png	Maturity stage effects
4	04_Q3_seasonal_effects.png	Seasonal effects
5	05_Q4_environmental_effects.png	Environmental modulation
6	06_Q5_uninfected_environment.png	Baseline environmental changes
7	07_Q6_transcriptome_trajectory.png	Infection age trajectory
8-13	08-13_pathway_enrichment.png	Pathway enrichment (Q1-Q6)
14	14_master_pathway_landscape.png	Master pathway landscape
15-19	15-19_seasonal_maturity_*.png	Seasonal maturity progression
20-22	20-22_biomarker_*.png	Biomarker discovery
23	23_analysis_schematic.png	Analysis pipeline flowchart
24-25	24-25_biomarker_QTL_*.png	GWAS/QTL cross-reference